

Matrices & Matrix Algebra



Matrix addition and scalar multiplication

- 1 Let A have rows $\langle 1, 2 \rangle, \langle 3, 4 \rangle$ and B have rows $\langle 0, -1 \rangle, \langle 2, 1 \rangle$. Find $A + B$ and $2A - B$.
 - 2 Let C have rows $\langle 2, 0, -1 \rangle, \langle 1, 3, 4 \rangle$. Find $-3C$.
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Matrix multiplication

- 3 Let A have rows $\langle 1, 2 \rangle, \langle 3, 4 \rangle$ and B have rows $\langle 2, 0 \rangle, \langle 1, 3 \rangle$. Find AB .
 - 4 For a 2×3 matrix A and a 3×2 matrix B , what are the dimensions of AB ? Is BA defined, and if so, what dimension?
 - 5 Let A have rows $\langle 1, 0, 2 \rangle, \langle -1, 3, 1 \rangle$ (a 2×3 matrix) and B have rows $\langle 2, 1 \rangle, \langle 0, 1 \rangle, \langle 3, -1 \rangle$ (a 3×2 matrix). Find AB .
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Matrix multiplication is not commutative

- 6 For A, B above (rows $\langle 1, 2 \rangle, \langle 3, 4 \rangle$ and $\langle 2, 0 \rangle, \langle 1, 3 \rangle$), compute BA and compare it to AB .
 - 7 Give one situation where $AB = BA$ IS guaranteed, even though matrix multiplication is not commutative in general.
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The identity matrix and matrix inverses

- 8 Describe the 2×2 and 3×3 identity matrices by their rows.
 - 9 Verify that $AI_2 = A$ for A with rows $\langle 3, 5 \rangle, \langle 2, 1 \rangle$.
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Invertibility of 2x2 matrices

- 10 For A with rows $\langle a, b \rangle, \langle c, d \rangle$, state the formula for A^{-1} .
- 11 Find the inverse of A with rows $\langle 3, 1 \rangle, \langle 2, 1 \rangle$.
- 12 Find the inverse of A with rows $\langle 2, 5 \rangle, \langle 1, 3 \rangle$.

- 13 Use the inverse found above to solve the system $2x + 5y = 7$, $x + 3y = 4$ (writing it as $A\mathbf{v} = \mathbf{b}$ and computing $\mathbf{v} = A^{-1}\mathbf{b}$).
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Inverse of a product

- 14 State the formula for $(AB)^{-1}$ in terms of A^{-1} and B^{-1} .
- 15 If A^{-1} has rows $\langle 2, 0 \rangle$, $\langle 1, 1 \rangle$ and B^{-1} has rows $\langle 1, 1 \rangle$, $\langle 0, 1 \rangle$, find $(AB)^{-1}$.
- 16 Explain briefly, using $(AB)(B^{-1}A^{-1}) = I$, why the inverse of a product must reverse the order.
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The transpose

- 17 For A with rows $\langle 1, 2, 3 \rangle$, $\langle 4, 5, 6 \rangle$ (a 2×3 matrix), find A^T .
- 18 State the formula for $(AB)^T$ in terms of A^T and B^T .
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Symmetric matrices

- 19 A matrix A is symmetric if $A^T = A$. Is the matrix with rows $\langle 2, 3 \rangle$, $\langle 3, 5 \rangle$ symmetric?
- 20 For any matrix A , explain briefly why AA^T is always symmetric, even when A itself is not square.
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Diagonal and triangular matrices

- 21 A diagonal matrix has rows $\langle 3, 0, 0 \rangle$, $\langle 0, -2, 0 \rangle$, $\langle 0, 0, 5 \rangle$. Find its inverse (recall: the inverse of a diagonal matrix is diagonal, with reciprocal entries).
- 22 Multiply two diagonal matrices D_1 (rows $\langle 2, 0 \rangle$, $\langle 0, 3 \rangle$) and D_2 (rows $\langle 5, 0 \rangle$, $\langle 0, -1 \rangle$). What pattern do you notice?
- 23 Explain why an upper triangular matrix (all entries below the main diagonal are zero) with a 0 on its main diagonal cannot be invertible.
- 24 Is the sum of two upper triangular matrices always upper triangular? Is the sum of two symmetric matrices always symmetric? Justify each briefly.

- 25** Find the inverse of the upper triangular matrix with rows $\langle 2, 3 \rangle$, $\langle 0, 4 \rangle$ using the 2×2 inverse formula, and confirm it is also upper triangular.